



FATIMA JINNAH WOMEN UNIVERSITY

RAWALPINDI

Department Of Software Engineering

Cloud Computing Lab

Saira Ejaz [2023-BSE-057]

Section: 5-B

LAB NO 5

Lab Task:

Task 1 - Discover missing command & install Java using apt suggestion

Goal: Run the java command when it's not installed, follow the system suggestion to install a Java package using apt, verify, then remove and clear shell cache.

Steps (inside the VM terminal)

1. Run the java command to see what the system suggests:

java

Expected: message similar to "The program 'java' can be found in the following packages: ...

Try: `sudo apt install "`

```
saira@ubuntu1804:~$ java
Command 'java' not found, but can be installed with:
sudo apt install default-jre          # version 2:1.17-75, or
sudo apt install openjdk-17-jre-headless # version 17.0.16+8~us1-0ubuntu1~24.04.1
sudo apt install openjdk-21-jre-headless # version 21.0.8+9~us1-0ubuntu1~24.04.1
sudo apt install openjdk-19-jre-headless # version 19.0.2+7-4
sudo apt install openjdk-20-jre-headless # version 20.0.2+9-1
sudo apt install openjdk-22-jre-headless # version 22~22ea-1
sudo apt install openjdk-11-jre-headless # version 11.0.28+6-1ubuntu1~24.04.1
sudo apt install openjdk-25-jre-headless # version 25+36-1~24.04.2
sudo apt install openjdk-8-jre-headless  # version 8u462-ga~us1-0ubuntu2~24.04.2
saira@ubuntu1804:~$
```

2. Use the suggested apt command (copy the exact package name suggested by the system). Example (replace with the name shown on your VM):

sudo apt install <suggested-package> -y

```
saira@ubuntu1804:~$ sudo apt install default-jre -y
[sudo] password for saira:
Sorry, try again.
[sudo] password for saira:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates-java dconf-gsettings-backend dconf-service
default-jre-headless fontconfig fonts-dejavu-extra gsettings-desktop-schemas gtk-update-icon-cache hicolor-icon-theme humanity-icon-theme ja
libasound2-data libasound2t64 libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0t64 libatspi2.0-0t64 libavahi-cl
libavahi-common-data libavahi-common3 libcairo-gobject2 libcairo2 libcolord2 libcups2t64 libdatrie1 libdconf1 libdrm-amdgpu1 libdrm-intel1 l
libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7 libgl1 libgl1-mesa-dri libglvnd0 libglx-mesa0 libglx0 libgraphite2-
libgtk-3-bin libgtk-3-common libharfbuzz0b libice6 liblcms2-2 liblvm20 libpango-1.0-0 libpangocairo-1.0-0 libpangoft2-1.0-0 libpciaccess0 l
libpixman-1-0 librsvg2-2 librsvg2-common libsm6 libthai-data libthai0 libvulkan1 libwayland-client0 libwayland-cursor0 libwayland-egl1 libwa
libx11-xcb1 libxaw7 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-render0 libxcb-shape0 libxcb-shm0 libxcb-sync1 libxcb-xfi
libxcomposite1 libxcursor1 libxdamage1 libxfixes3 libxft2 libxi6 libxinerama1 libxkbfile1 libxmu6 libxrandr2 libxrender1 libxshmfence1 libxt
libxv1 libxxf86dga1 libxxf86vm1 mesa-libgallium mesa-vulkan-drivers openjdk-21-jre openjdk-21-jre-headless session-migration ubuntu-mono x11
x11-utils
Suggested packages:
alsa-utils libasound2-plugins colord cups-common gvfs liblcms2-utils pcsd librsvg2-bin libnss-mdns fonts-ipafont-gothic fonts-ipafont-minch
fonts-wqy-microhei | fonts-wqy-zenhei fonts-indic mesa-utils
Recommended packages:
luit
The following NEW packages will be installed:
adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates-java dconf-gsettings-backend dconf-service d
default-jre-headless fontconfig fonts-dejavu-extra gsettings-desktop-schemas gtk-update-icon-cache hicolor-icon-theme humanity-icon-theme ja
libasound2-data libasound2t64 libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0t64 libatspi2.0-0t64 libavahi-cl
libavahi-common-data libavahi-common3 libcairo-gobject2 libcairo2 libcolord2 libcups2t64 libdatrie1 libdconf1 libdrm-amdgpu1 libdrm-intel1 l
libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7 libgl1 libgl1-mesa-dri libglvnd0 libglx-mesa0 libglx0 libgraphite2-
libgtk-3-bin libgtk-3-common libharfbuzz0b libice6 liblcms2-2 liblvm20 libpango-1.0-0 libpangocairo-1.0-0 libpangoft2-1.0-0 libpciaccess0 l
libpixman-1-0 librsvg2-2 librsvg2-common libsm6 libthai-data libthai0 libvulkan1 libwayland-client0 libwayland-cursor0 libwayland-egl1 libwa
libx11-xcb1 libxaw7 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-render0 libxcb-shape0 libxcb-shm0 libxcb-sync1 libxcb-xfi
libxcomposite1 libxcursor1 libxdamage1 libxfixes3 libxft2 libxi6 libxinerama1 libxkbfile1 libxmu6 libxrandr2 libxrender1 libxshmfence1 libxt
libxv1 libxxf86dga1 libxxf86vm1 mesa-libgallium mesa-vulkan-drivers openjdk-21-jre openjdk-21-jre-headless session-migration ubuntu-mono x11
x11-utils
0 upgraded, 106 newly installed, 0 to remove and 64 not upgraded.
```

```
Selecting previously unselected package libdconf1:amd64.
Preparing to unpack .../016-libdconf1_0.40.0-4ubuntu0.1_amd64.deb ...
Unpacking libdconf1:amd64 (0.40.0-4ubuntu0.1) ...
Selecting previously unselected package dconf-service.
Preparing to unpack .../017-dconf-service_0.40.0-4ubuntu0.1_amd64.deb ...
Unpacking dconf-service (0.40.0-4ubuntu0.1) ...
Selecting previously unselected package dconf-gsettings-backend:amd64.
Preparing to unpack .../018-dconf-gsettings-backend_0.40.0-4ubuntu0.1_amd64.deb ...
Unpacking dconf-gsettings-backend:amd64 (0.40.0-4ubuntu0.1) ...
Selecting previously unselected package session-migration.
Preparing to unpack .../019-session-migration_0.3.9build1_amd64.deb ...
Unpacking session-migration (0.3.9build1) ...
Selecting previously unselected package gsettings-desktop-schemas.
Preparing to unpack .../020-gsettings-desktop-schemas_46.1-0ubuntu1_all.deb ...
Unpacking gsettings-desktop-schemas (46.1-0ubuntu1) ...
Selecting previously unselected package at-spi2-core.
Preparing to unpack .../021-at-spi2-core_2.52.0-1build1_amd64.deb ...
Unpacking at-spi2-core (2.52.0-1build1) ...
Selecting previously unselected package ca-certificates-java.
Preparing to unpack .../022-ca-certificates-java_20240118_all.deb ...
Unpacking ca-certificates-java (20240118) ...
Selecting previously unselected package java-common.
Preparing to unpack .../023-java-common_0.75+exp1_all.deb ...
Unpacking java-common (0.75+exp1) ...
Selecting previously unselected package liblcms2-2:amd64.
Preparing to unpack .../024-liblcms2-2_2.14-2build1_amd64.deb ...
Unpacking liblcms2-2:amd64 (2.14-2build1) ...
Selecting previously unselected package libpcsclite1:amd64.
Preparing to unpack .../025-libpcsclite1_2.0.3-1build1_amd64.deb ...
Unpacking libpcsclite1:amd64 (2.0.3-1build1) ...
Selecting previously unselected package openjdk-21-jre-headless:amd64.
Preparing to unpack .../026-openjdk-21-jre-headless_21.0.8+9~us1-0ubuntu1~24.04.1_amd64.deb ...
Unpacking openjdk-21-jre-headless:amd64 (21.0.8+9~us1-0ubuntu1~24.04.1) ...
```

Progress: [12%] [#####.....]

3. Verify Java is installed and check version:

java -version

```

No VM guests are running outdated hypervisor (qemu) binaries on this host.
saira@ubuntu:~$ java --version
openjdk 21.0.8 2025-07-15
OpenJDK Runtime Environment (build 21.0.8+9-Ubuntu-0ubuntu124.04.1)
OpenJDK 64-Bit Server VM (build 21.0.8+9-Ubuntu-0ubuntu124.04.1, mixed mode, sharing)
saira@ubuntu:~$ _

```

4. Remove the Java package using apt remove (use the same package name you installed):

sudo apt remove <suggested-package> -y

```

saira@ubuntu:~$ sudo apt remove default-jre -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates-java dconf-gsettings-backend dconf-service
default-jre-headless fontconfig fonts-dejavu-extra gsettings-desktop-schemas gtk-update-icon-cache hicolor-icon-theme humanity-icon-theme java-common
libasound2-data libasound2t64 libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0t64 libatspi2.0-0t64 libavahi-client3
libavahi-common-data libavahi-common3 libcairo-gobject2 libcairo2 libcolord2 libcups2t64 libdatrie1 libdconf1 libdrm-amdgpu1 libdrm-intel1 libepoxy0 libgbm1
libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgl1 libgl1-mesa-dri libglvnd0 libglx-mesa0 libglx0 libgraphite2-3 libgtk-3-0t64
libgtk-3-bin libgtk-3-common libharfbuzz0b libice6 liblcms2-2 liblvm20 libpango-1.0-0 libpangocairo-1.0-0 libpangoft2-1.0-0 libpciaccess0 libpcsclite1
libpixman-1-0 librsvg2-2 librsvg2-common libsm6 libthai-data libthai0 libvulkan1 libwayland-client0 libwayland-cursor0 libwayland-egl1 libwayland-server0
libx11-xcb1 libxaw7 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-render0 libxcb-shape0 libxcb-shm0 libxcb-sync1 libxcb-xfixes0
libxcomposite1 libxcursor1 libxdamage1 libxfixes3 libxft2 libxi6 libxinerama1 libxkbfile1 libxmu6 libxrandr2 libxrender1 libxshmfence1 libxt6t64 libxtst6
libxv1 libxxf86dga1 libxxf86vm1 mesa-libgallium mesa-vulkan-drivers openjdk-21-jre openjdk-21-jre-headless session-migration ubuntu-mono x11-common
x11-utils
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
default-jre
0 upgraded, 0 newly installed, 1 to remove and 65 not upgraded.
After this operation, 6,144 B disk space will be freed.
(Reading database ... 101562 files and directories currently installed.)
Removing default-jre (2:1.21-75+exp1) ...
saira@ubuntu:~$

```

5. Confirm java is no longer available (run java again) it should again indicate "not found" or suggest installation:

Java

```

        set a system property
-verbose:[class|module|gc|jni]
        enable verbose output for the given subsystem
-version
        print product version to the error stream and exit
--version
        print product version to the output stream and exit
-showversion
        print product version to the error stream and continue
--show-version
        print product version to the output stream and continue
--show-module-resolution
        show module resolution output during startup
-? -h -help
        print this help message to the error stream
--help
        print this help message to the output stream
-X
        print help on extra options to the error stream
--help-extra
        print help on extra options to the output stream
-ea[:<packagename>...[:<classname>]]
-enableassertions[:<packagename>...[:<classname>]]
        enable assertions with specified granularity
-da[:<packagename>...[:<classname>]]
-disableassertions[:<packagename>...[:<classname>]]
        disable assertions with specified granularity
-esa | -enablesystemassertions
        enable system assertions
-dsa | -disablesystemassertions
        disable system assertions
-agentlib:<libname>[=<options>]
        load native agent library <libname>, e.g. -agentlib:jdwp
        see also -agentlib:jdwp=help
-agentpath:<pathname>[=<options>]
        load native agent library by full pathname
-javaagent:<jarpath>[=<options>]
        load Java programming language agent, see java.lang.instrument
-splash:<imagepath>
        show splash screen with specified image
        HiDPI scaled images are automatically supported and used
        if available. The unscaled image filename, e.g. image.ext,
        should always be passed as the argument to the -splash option.
        The most appropriate scaled image provided will be picked up
        automatically.
        See the SplashScreen API documentation for more information
@argument files
        one or more argument files containing options
--disable-@files
        prevent further argument file expansion
--enable-preview
        allow classes to depend on preview features of this release
To specify an argument for a long option, you can use --<name>=<value> or
--<name> <value>.

saira@ubuntu:~$

```

6. Clear the shell's command hash cache so the shell forgets cached command locations (run as your regular user — no sudo required):

hash -r

java

```

(reading database ... 101582 files and directories currently installed.)
Removing default-jre (2:1.21-75+exp1) ...
saira@ubuntu:~$ hash -r
saira@ubuntu:~$ java

```

```

        set a system property
-verbose:[class|module|gc|jni]
        enable verbose output for the given subsystem
-version          print product version to the error stream and exit
--version         print product version to the output stream and exit
-showversion      print product version to the error stream and continue
--show-version    print product version to the output stream and continue
--show-module-resolution
        show module resolution output during startup
-? -h -help       print this help message to the error stream
--help            print this help message to the output stream
-X               print help on extra options to the error stream
--help-extra      print help on extra options to the output stream
-ea[:<packagename>...|:<classname>]
-enableassertions[:<packagename>...|:<classname>]
        enable assertions with specified granularity
-da[:<packagename>...|:<classname>]
-disableassertions[:<packagename>...|:<classname>]
        disable assertions with specified granularity
-esa | -enablesystemassertions
        enable system assertions
-dsa | -disablesystemassertions
        disable system assertions
-agentlib:<libname>[=<options>]
        load native agent library <libname>, e.g. -agentlib:jdwp
        see also -agentlib:jdwp=help
-agentpath:<pathname>[=<options>]
        load native agent library by full pathname
-javaagent:<jarpath>[=<options>]
        load Java programming language agent, see java.lang.instrument
-splash:<imagepath>
        show splash screen with specified image
        HiDPI scaled images are automatically supported and used
        if available. The unscaled image filename, e.g. image.ext,
        should always be passed as the argument to the -splash option.
        The most appropriate scaled image provided will be picked up
        automatically.
        See the SplashScreen API documentation for more information
@argument files
        one or more argument files containing options
--disable-@files
        prevent further argument file expansion
--enable-preview
        allow classes to depend on preview features of this release
To specify an argument for a long option, you can use --<name>=<value> or
--<name> <value>.

saira@ubuntu:~$

```

Task 2 - Install & remove Java using apt-get (explicitly)

Goal: Repeat install/remove using the apt-get tool to show both package managers.

Steps (inside VM terminal)

1. Install Java using apt-get (choose a common package, e.g., default-jre — or the same package you used in Task 1):

sudo apt-get update

```
saira@ubuntu:~$ sudo apt-get update

Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
saira@ubuntu:~$ _
```

sudo apt-get install default-jre -y

```
Reading package lists... Done
saira@ubuntu:~$ sudo apt-get install default-jre -y

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  default-jre
0 upgraded, 1 newly installed, 0 to remove and 65 not upgraded.
Need to get 922 B of archives.
After this operation, 6,144 B of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 default-jre amd64 2:1.21-75+exp1 [922 B]
Fetched 922 B in 1s (709 B/s)
Selecting previously unselected package default-jre.
(Reading database ... 101561 files and directories currently installed.)
Preparing to unpack .../default-jre_2%3a1.21-75+exp1_amd64.deb ...
Unpacking default-jre (2:1.21-75+exp1) ...
Setting up default-jre (2:1.21-75+exp1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

2. Verify Java version again:

java -version

```
saira@ubuntu:~$ java --version
openjdk 21.0.8 2025-07-15
OpenJDK Runtime Environment (build 21.0.8+9-Ubuntu-0ubuntu124.04.1)
OpenJDK 64-Bit Server VM (build 21.0.8+9-Ubuntu-0ubuntu124.04.1, mixed mode, sharing)
saira@ubuntu:~$ _
```

3. Remove Java using apt-get remove:

sudo apt-get remove default-jre -y

```

openjdk-8-jdk-server in 100110-0ubuntu1; ubuntu: ubuntu:14.04.1; mixed mode; sharing)
saira@ubuntu:~$ sudo apt-get remove default-jre -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates-java dconf-gsettings-backend dconf-service
  default-jre-headless fontconfig fonts-dejavu-extra gsettings-desktop-schemas gtk-update-icon-cache hicolor-icon-theme humanity-icon-theme java-common
  libasound2-data libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0t64 libatspi2.0-0t64 libavahi-client3
  libavahi-common-data libavahi-common3 libcairo-gobject2 libcairo2 libcolor2 libcolor2 libcupst64 libdatrie1 libdconf1 libdrm-amdgpu1 libdrm-intel1 libepoxy0 libgbm1
  libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgl1 libgl1-mesa-dri libglvnd0 libglx-mesa0 libglx0 libgraphite2-3 libgtk-3-0t64
  libgtk-3-bin libgtk-3-common libharfbuzz0b libice6 liblcms2-2 liblvm20 libpango-1.0-0 libpangocairo-1.0-0 libpangoft2-1.0-0 libpciaccess0 libpcsclite1
  libpixman-1-0 librsvg2-2 librsvg2-common libsm6 libthai-data libthai0 libvulkan1 libwayland-client0 libwayland-cursor0 libwayland-egl1 libwayland-server0
  libx11-xcb1 libxaw7 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-render0 libxcb-shape0 libxcb-shm0 libxcb-sync1 libxcb-xf86vm0
  libxcomposite1 libxcursor1 libxdamage1 libxfixes3 libxft2 libxi6 libxinerama1 libxkbfile1 libxmu6 libxrandr2 libxrender1 libxshmfence1 libxt6t64 libxtst6
  libxxf86dga1 libxxf86vm1 mesa-libgallium mesa-vulkan-drivers openjdk-21-jre openjdk-21-jre-headless session-migration ubuntu-mono x11-common
  x11-utils
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  default-jre
0 upgraded, 0 newly installed, 1 to remove and 65 not upgraded.
After this operation, 6,144 B disk space will be freed.
(Reading database ... 101562 files and directories currently installed.)
Removing default-jre (2:1.21-75+exp1) ...
saira@ubuntu:~$

```

4. Clear the terminal hash cache and confirm java is missing:

hash -r

java

```

(Reading database ... 101562 files and directories currently installed.)
Removing default-jre (2:1.21-75+exp1) ...
saira@ubuntu:~$ hash -r
saira@ubuntu:~$ java

```

```

        set a system property
-verbose:[class|module|gc|jni]
    enable verbose output for the given subsystem
-version
    print product version to the error stream and exit
--version
    print product version to the output stream and exit
-showversion
    print product version to the error stream and continue
--show-version
    print product version to the output stream and continue
--show-module-resolution
    show module resolution output during startup
-? -h -help
    print this help message to the error stream
--help
    print this help message to the output stream
-X
    print help on extra options to the error stream
--help-extra
    print help on extra options to the output stream
-ea[:<packagename>...[:<classname>]]
-enableassertions[:<packagename>...[:<classname>]]
    enable assertions with specified granularity
-da[:<packagename>...[:<classname>]]
-disableassertions[:<packagename>...[:<classname>]]
    disable assertions with specified granularity
-esa | -enablesystemassertions
    enable system assertions
-dsa | -disablesystemassertions
    disable system assertions
-agentlib:<libname>[=<options>]
    load native agent library <libname>, e.g. -agentlib:jdwp
    see also -agentlib:jdwp=help
-agentpath:<pathname>[=<options>]
    load native agent library by full pathname
-javaagent:<jarpath>[=<options>]
    load Java programming language agent, see java.lang.instrument
-splash:<imagepath>
    show splash screen with specified image
    HiDPI scaled images are automatically supported and used
    if available. The unscaled image filename, e.g. image.ext,
    should always be passed as the argument to the -splash option.
    The most appropriate scaled image provided will be picked up
    automatically.
    See the SplashScreen API documentation for more information
@argument files
    one or more argument files containing options
--disable-@files
    prevent further argument file expansion
--enable-preview
    allow classes to depend on preview features of this release
To specify an argument for a long option, you can use --<name>=<value> or
--<name> <value>.

saira@ubuntu:~$ _

```

Task 3 - apt update vs apt upgrade - run & explain

Goal: Run the commands and document the difference.

Steps (inside VM terminal)

1. Update the package index (this downloads the latest lists of available packages):

sudo apt update


```
saira@ubuntu:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
65 packages can be upgraded. Run 'apt list --upgradable' to see them.
saira@ubuntu:~$
```

2. Upgrade installed packages (this installs available updates for currently installed packages):

sudo apt upgrade

```
Get:29 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libpython3.12-minimal amd64 3.12.3-1ubuntu0.8 [836 kB]
Get:30 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libsqlite3-0 amd64 3.45.1-1ubuntu2.5 [701 kB]
Get:31 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 ubuntu-drivers-common amd64 1:0.9.7.6ubuntu3.4 [65.8 kB]
Get:32 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libxml2 amd64 2.9.14+dfsg-1.3ubuntu3.6 [763 kB]
Get:33 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 open-vm-tools amd64 2:12.5.0-1ubuntu0.24.04.2 [742 kB]
Get:34 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 distro-info-data all 0.60ubuntu0.5 [6,934 B]
Get:35 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 locales all 2.39-0ubuntu8.6 [4,229 kB]
Get:36 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 openssl amd64 3.0.13-0ubuntu3.6 [1,003 kB]
Get:37 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 systemd-hwe-hwdb all 255.1.6 [3,604 B]
Get:38 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 vim amd64 2:9.1.0016-1ubuntu7.9 [1,081 kB]
Get:39 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 vim-common all 2:9.1.0016-1ubuntu7.9 [386 kB]
Get:40 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 vim-tiny amd64 2:9.1.0016-1ubuntu7.9 [809 kB]
Get:41 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 vim-runtime all 2:9.1.0016-1ubuntu7.9 [7,281 kB]
Get:42 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 xxd amd64 2:9.1.0016-1ubuntu7.9 [63.8 kB]
Get:43 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-dnsutils amd64 1:9.18.39-0ubuntu0.24.04.2 [156 kB]
Get:44 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-host amd64 1:9.18.39-0ubuntu0.24.04.2 [50.5 kB]
Get:45 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-libs amd64 1:9.18.39-0ubuntu0.24.04.2 [1,257 kB]
Get:46 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 powermgmt-base all 1.37ubuntu0.1 [7,650 B]
Get:47 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 tcpdump amd64 4.99.4-3ubuntu4.24.04.1 [479 kB]
Get:48 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 firmware-sof-signed all 2023.12.1-1ubuntu1.10 [7,382 kB]
Get:49 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libfwupd2 amd64 1.9.31-0ubuntu1~24.04.1 [136 kB]
Get:50 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 fwupd amd64 1.9.31-0ubuntu1~24.04.1 [4,592 kB]
Ign:50 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 fwupd amd64 1.9.31-0ubuntu1~24.04.1
Get:51 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 landscape-common amd64 24.02-0ubuntu5.6 [93.2 kB]
Get:52 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libssh-4 amd64 0.10.6-2ubuntu0.2 [188 kB]
Get:53 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libtirpc amd64 4.5.1+git230720-4ubuntu2.4 [199 kB]
Get:54 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 udisk2 amd64 2.10.1-6ubuntu1.3 [297 kB]
Get:55 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libudisks2-0 amd64 2.10.1-6ubuntu1.3 [175 kB]
Get:56 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-firmware amd64 20240318.git3b128b60-0ubuntu2.19 [538 MB]
Ign:56 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-firmware amd64 20240318.git3b128b60-0ubuntu2.19
Get:57 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-modules-6.8.0-87-generic amd64 6.8.0-87.88 [39.4 MB]
Get:58 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-image-6.8.0-87-generic amd64 6.8.0-87.88 [14.8 MB]
Get:59 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-modules-extra-6.8.0-87-generic amd64 6.8.0-87.88 [113 MB]
Get:60 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-generic amd64 6.8.0-87.88 [1,702 B]
Get:61 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-image-generic amd64 6.8.0-87.88 [11.0 kB]
Get:62 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-headers-6.8.0-87 all 6.8.0-87.88 [14.0 MB]
Get:63 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-headers-6.8.0-87-generic amd64 6.8.0-87.88 [4,163 kB]
Get:64 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-headers-generic amd64 6.8.0-87.88 [10.8 kB]
Get:65 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-common all 6.8.0-87.88 [772 kB]
Get:66 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-6.8.0-87 amd64 6.8.0-87.88 [5,432 kB]
Get:67 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-tools-6.8.0-87-generic amd64 6.8.0-87.88 [1,810 B]
Get:68 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 software-properties-common all 0.99.49.3 [14.4 kB]
Get:69 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 python3-software-properties all 0.99.49.3 [29.9 kB]
Get:70 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 snapd amd64 2.71+ubuntu24.04 [33.9 MB]
Ign:70 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 snapd amd64 2.71+ubuntu24.04
Get:71 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 sosreport amd64 4.9.2-0ubuntu0~24.04.1 [372 kB]
Get:72 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 cloud-init all 25.2-0ubuntu1~24.04.1 [625 kB]
Get:50 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 fwupd amd64 1.9.31-0ubuntu1~24.04.1 [4,592 kB]
Get:56 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 linux-firmware amd64 20240318.git3b128b60-0ubuntu2.19 [538 MB]
70% [56 linux-firmware 271 MB/538 MB 50%]
```

744 kB/s 6min 43s_

```
Preconfiguring packages ...
(Reading database ... 101561 files and directories currently installed.)
Preparing to unpack .../coreutils_9.4-3ubuntu6.1_amd64.deb ...
Unpacking coreutils (9.4-3ubuntu6.1) over (9.4-3ubuntu6) ...
Setting up coreutils (9.4-3ubuntu6.1) ...
(Reading database ... 101561 files and directories currently installed.)
Preparing to unpack .../dpkg_1.22.6ubuntu6.5_amd64.deb ...
Unpacking dpkg (1.22.6ubuntu6.5) over (1.22.6ubuntu6.1) ...
Setting up dpkg (1.22.6ubuntu6.5) ...
(Reading database ... 101561 files and directories currently installed.)
Preparing to unpack .../libc-devtools_2.39-0ubuntu8.6_amd64.deb ...
Unpacking libc-devtools (2.39-0ubuntu8.6) over (2.39-0ubuntu8.5) ...
Preparing to unpack .../libc6-dev_2.39-0ubuntu8.6_amd64.deb ...
Unpacking libc6-dev:amd64 (2.39-0ubuntu8.6) over (2.39-0ubuntu8.5) ...
Preparing to unpack .../libc-dev-bin_2.39-0ubuntu8.6_amd64.deb ...
Unpacking libc-dev-bin (2.39-0ubuntu8.6) over (2.39-0ubuntu8.5) ...
Preparing to unpack .../linux-libc-dev_6.8.0-87.88_amd64.deb ...
Unpacking linux-libc-dev:amd64 (6.8.0-87.88) over (6.8.0-71.71) ...
Preparing to unpack .../libc6_2.39-0ubuntu8.6_amd64.deb ...
Unpacking libc6:amd64 (2.39-0ubuntu8.6) over (2.39-0ubuntu8.5) ...
Setting up libc6:amd64 (2.39-0ubuntu8.6) ...
(Reading database ... 101561 files and directories currently installed.)
Preparing to unpack .../libc-bin_2.39-0ubuntu8.6_amd64.deb ...
Unpacking libc-bin (2.39-0ubuntu8.6) over (2.39-0ubuntu8.5) ...
Setting up libc-bin (2.39-0ubuntu8.6) ...
(Reading database ... 101561 files and directories currently installed.)
Preparing to unpack .../libpam0g_1.5.3-5ubuntu5.5_amd64.deb ...
Unpacking libpam0g:amd64 (1.5.3-5ubuntu5.5) over (1.5.3-5ubuntu5.4) ...
Setting up libpam0g:amd64 (1.5.3-5ubuntu5.5) ...
(Reading database ... 101561 files and directories currently installed.)
Preparing to unpack .../systemd-timesyncd_255.4-1ubuntu8.11_amd64.deb ...
Unpacking systemd-timesyncd (255.4-1ubuntu8.11) over (255.4-1ubuntu8.10) ...
Preparing to unpack .../libssl1.1_3.0.13-0ubuntu3.6_amd64.deb ...
Unpacking libssl1.1:amd64 (3.0.13-0ubuntu3.6) over (3.0.13-0ubuntu3.5) ...

Progress: [ 11%] [#####.....]
```

```

Processing triggers for systemd (255.4-1ubuntu8.11) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for install-info (7.1-3build2) ...
Processing triggers for linux-image-6.8.0-87-generic (6.8.0-87.88) ...
/etc/kernel/postinst.d/initramfs-tools:
update-initramfs: Generating /boot/initrd.img-6.8.0-87-generic
/etc/kernel/postinst.d/zz-update-grub:
Sourcing file `/etc/default/grub'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-6.8.0-87-generic
Found initrd image: /boot/initrd.img-6.8.0-87-generic
Found linux image: /boot/vmlinuz-6.8.0-71-generic
Found initrd image: /boot/initrd.img-6.8.0-71-generic
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
 6.8.0-71-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-87-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...
systemctl restart cron.service multipathd.service polkit.service ssh.service upower.service

Service restarts being deferred:
systemctl restart ModemManager.service
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
saira @ session #10: sshd[3520,3576]
saira @ session #11: apt[6174], sshd[3592,3648]
saira @ session #2: bash[1525], login[1045]
saira @ user manager service: systemd[1513]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
saira@ubuntu:~$

```

- Write a short 3–5 sentence explanation describing the difference between apt update and apt upgrade. Put your text into a small file and capture it as a screenshot (do not upload the text file; provide the screenshot):

nano ~/apt_update_vs_upgrade.md

write 3-5 sentences, save and exit

```

GNU nano 7.2 /home/saira/apt_update_vs_upgrade.md *
The command 'sudo apt update' is used to refresh the package index, which means it downloads the latest information about available packages and their versions.
The command 'sudo apt upgrade' actually installs the new versions of packages that are already installed on the system if updates are available.
In short, 'apt update' prepares the system for upgrades, while 'apt upgrade' performs the actual updates.
Both are often run together to keep the system up to date.

```

```
saira@ubuntu:~$ cat /home/saira/apt_update_vs_upgrade.md
```

```
The command 'sudo apt update' is used to refresh the package index, which means it downloads the latest information about available packages and their versions.
The command 'sudo apt upgrade' actually installs the new versions of packages that are already installed on the system if updates are available.
In short, 'apt update' prepares the system for upgrades, while 'apt upgrade' performs the actual updates.
Both are often run together to keep the system up to date.
```

```
saira@ubuntu:~$
```

Task 4 - Install Visual Studio Code via snap on CLI and verify (DO NOT remove Code)

Goal: Install a snap package (example: VS Code) on a CLI-only server and verify installation using commands that work without a GUI. Keep Code installed for later GUI tasks.

Steps (inside VM terminal)

1. Install VS Code via snap (snap may require sudo):

sudo snap install --classic code

```
saira@ubuntu:~$ sudo snap install --classic code
[sudo] password for saira:
Download snap "snapd" (25577) from channel "stable"
```

19% 1.14MB/s 38.1s

```
saira@ubuntu:~$ sudo snap install --classic code
[sudo] password for saira:
2025-11-04T10:36:26Z INFO Waiting for automatic snapd restart...
code 7d842fb8 from Visual Studio Code (vscode) installed
saira@ubuntu:~$
```

2. Verify snap shows the package is installed:

snap list code

```
saira@ubuntu:~$ snap list code
code 7d842fb8 from Visual Studio Code (vscode) installed
Name Version Rev Tracking Publisher Notes
code 7d842fb8 211 latest/stable vscode classic
saira@ubuntu:~$ _
```

3. Check the installed application's version. On some systems code --version is available; also check snap info:

code --version

```
saira@ubuntu:~$ code --version
1.105.1
7d842fb85a0275a4a8e4d7e040d2625abbf7f084
x64
saira@ubuntu:~$ _
```

4. If the code binary is not in PATH, show where the snap placed it:

ls -l /snap/bin | grep code

```
Warning: The unit file, source configuration file or drop-ins of keyboard-occur-
saira@ubuntu:~$ ls -l /snap/bin | grep code
lrwxrwxrwx 1 root root 13 Nov  4 10:44 code -> /usr/bin/snap
lrwxrwxrwx 1 root root 13 Nov  4 10:44 code.url-handler -> /usr/bin/snap
saira@ubuntu:~$
```

Task 5 - Install XFCE GUI + XRDP - minimal desktop and remote access (GUI) and launch VS Code

Goal: Install a lightweight GUI (XFCE), enable XRDP for remote desktop access, and learn how to control whether the GUI login screen appears at boot. After starting the GUI, launch the VS Code installed in Task 4 to verify GUI app launch. This task is for this lab.

Steps (inside the host terminal / via SSH)

1. From your host, open your preferred terminal (for example: Windows Command Prompt, PowerShell, macOS Terminal, or Linux Terminal) and connect to the VM using SSH. Example:

ssh student@<vm-ip-address>

```
saira@ubuntu:~$ ssh saira@192.168.154.128
saira@192.168.154.128's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-87-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue  4 Nov 10:53:18 UTC 2025

System load:  0.06               Processes:            219
Usage of /:   60.5% of 9.75GB    Users logged in:     1
Memory usage: 8%                IPv4 address for ens33: 192.168.154.128
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Nov  4 08:51:45 2025 from 192.168.154.128
saira@ubuntu:~$ _
```

2. Update the server (download package lists and apply upgrades):

sudo apt update && sudo apt upgrade -y

```
saira@ubuntu:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for saira:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:8 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,156 B]
Get:9 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:10 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.9 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 825 kB in 10s (84.9 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  ca-certificates-java default-jre-headless fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-jni libgif7 libice6 libpcsc-lite1 libsm6
  libxau7 libxcb-shape0 libxft2 libxkbfile1 libxmu6 libxft6 libxft6dgal openjdk-21-jre openjdk-21-jre-headless x11-utils
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
saira@ubuntu:~$
```

3. Install XFCE and XFCE goodies (lightweight desktop):

sudo apt install xfce4 xfce4-goodies -y

```
Get:311 http://archive.ubuntu.com/ubuntu noble/main amd64 gtk2-engines-murrine amd64 0.98.2-4 [106 kB]
Get:312 http://archive.ubuntu.com/ubuntu noble/universe amd64 greybird-gtk-theme all 3.23.3-1 [583 kB]
Get:313 http://archive.ubuntu.com/ubuntu noble/main amd64 gsettings-ubuntu-schemas all 0.0.7+21.10.20210712-0ubuntu3 [5,774 B]
Get:314 http://archive.ubuntu.com/ubuntu noble/universe amd64 libcogl20 amd64 1.22.8-4ubuntu4 [328 kB]
Get:315 http://archive.ubuntu.com/ubuntu noble/universe amd64 libcogl-pango20 amd64 1.22.8-4ubuntu4 [16.5 kB]
Get:316 http://archive.ubuntu.com/ubuntu noble/universe amd64 libcogl-path20 amd64 1.22.8-4ubuntu4 [33.5 kB]
Get:317 http://archive.ubuntu.com/ubuntu noble/main amd64 libmtdev1t64 amd64 1.1.6-1.1build1 [14.4 kB]
Get:318 http://archive.ubuntu.com/ubuntu noble/main amd64 libwacom-common all 2.10.0-2 [63.4 kB]
Get:319 http://archive.ubuntu.com/ubuntu noble/main amd64 libwacom9 amd64 2.10.0-2 [23.9 kB]
Get:320 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libinput-bin amd64 1.25.0-1ubuntu3.1 [22.6 kB]
Get:321 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libinput10 amd64 1.25.0-1ubuntu3.1 [133 kB]
Get:322 http://archive.ubuntu.com/ubuntu noble/universe amd64 libclutter-1.0-0 amd64 1.26.4+git2779b932+dfsg-7build3 [566 kB]
Get:323 http://archive.ubuntu.com/ubuntu noble/universe amd64 libclutter-gst-3.0-0 amd64 3.0.27-4build1 [60.6 kB]
Get:324 http://archive.ubuntu.com/ubuntu noble/universe amd64 gstreamer1.0-clutter-3.0 amd64 3.0.27-4build1 [6,600 B]
Get:325 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gstreamer1.0-gl amd64 1.24.2-1ubuntu0.3 [116 kB]
Get:326 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gstreamer1.0-x amd64 1.24.2-1ubuntu0.3 [85.1 kB]
Get:327 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gvfs-common all 1.54.4-0ubuntu1~24.04.1 [13.7 kB]
Get:328 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gvfs-libs amd64 1.54.4-0ubuntu1~24.04.1 [107 kB]
Get:329 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gvfs-daemons amd64 1.54.4-0ubuntu1~24.04.1 [116 kB]
Get:330 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 gvfs amd64 1.54.4-0ubuntu1~24.04.1 [116 kB]
Get:331 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcdio19t64 amd64 2.1.0-4.1ubuntu1.2 [62.4 kB]
Get:332 http://archive.ubuntu.com/ubuntu noble/main amd64 libcdio-cdda2t64 amd64 10.2+2.0.1-1.1build2 [16.5 kB]
Get:333 http://archive.ubuntu.com/ubuntu noble/main amd64 libcdio-paranoia2t64 amd64 10.2+2.0.1-1.1build2 [16.6 kB]
Get:334 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libsoup2.4-common all 2.74.3-6ubuntu1.6 [9,372 B]
Get:335 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libsoup-2.4-1 amd64 2.74.3-6ubuntu1.6 [283 kB]
Get:336 http://archive.ubuntu.com/ubuntu noble/main amd64 libgdata-common all 0.18.1-6build2 [3,816 B]
Get:337 http://archive.ubuntu.com/ubuntu noble/main amd64 libgdata22 amd64 0.18.1-6build2 [284 kB]
Get:338 http://archive.ubuntu.com/ubuntu noble/main amd64 libmsgrahp-0-1 amd64 0.2.1-0ubuntu3 [31.1 kB]
Get:339 http://archive.ubuntu.com/ubuntu noble/main amd64 libmt0-common all 1.1.21-3.1build1 [19.6 kB]
Get:340 http://archive.ubuntu.com/ubuntu noble/main amd64 libmt0t64 amd64 1.1.21-3.1build1 [162 kB]
Get:341 http://archive.ubuntu.com/ubuntu noble/main amd64 libnfs14 amd64 5.0.2-1build1 [109 kB]
Get:342 http://archive.ubuntu.com/ubuntu noble/main amd64 libtalloc2 amd64 2.4.2-1build2 [27.3 kB]
Get:343 http://archive.ubuntu.com/ubuntu noble/main amd64 libtevent0t64 amd64 0.16.1-2build1 [42.6 kB]
Get:344 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libldb2 amd64 2:2.8.0+samba4.19.5+dfsg-4ubuntu9.4 [188 kB]
Get:345 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libwbclient0 amd64 2:4.19.5+dfsg-4ubuntu9.4 [71.7 kB]
Get:346 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 samba-libs amd64 2:4.19.5+dfsg-4ubuntu9.4 [6,018 kB]
53% [346 samba-libs 3,091 kB/6,018 kB 51%]
```

```

Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Setting up unity-gtk2-module:amd64 (0.0.0+18.04.20171202-0ubuntu7) ...
Setting up libnm0:amd64 (1.10.6-3build2) ...
Setting up light-locker-settings (1.5.0-0ubuntu4) ...
/usr/share/light-locker-settings/light-locker-settings/light-locker-settings.py:514: SyntaxWarning: invalid escape sequence '\d'
  "timeout: *(\d+)", screensaver_output)
/usr/share/light-locker-settings/light-locker-settings/light-locker-settings.py:517: SyntaxWarning: invalid escape sequence '\d'
  '\d+', screenblank_timeout_grep.group(1))
/usr/share/light-locker-settings/light-locker-settings/light-locker-settings.py:522: SyntaxWarning: invalid escape sequence '\d'
  "Standby: *(\d+)", screensaver_output)
/usr/share/light-locker-settings/light-locker-settings/light-locker-settings.py:525: SyntaxWarning: invalid escape sequence '\d'
  '\d+', screenoff_timeout_grep.group(1))
Setting up xiccd (0.3.0-2build2) ...
Setting up libadwaita-1-0:amd64 (1.5.0-1ubuntu2) ...
Setting up indicator-sound (12.10.2+18.10.20180612-0ubuntu7) ...
Setting up gkbd-capplet (3.28.1-1build3) ...
Setting up gnome-power-manager (43.0-2build2) ...
Setting up gvfs-backends (1.54.4-0ubuntu1~24.04.1) ...
Setting up libebook-contacts-1.2-4t64:amd64 (3.52.3-0ubuntu1) ...
Setting up libcheese-gtk25:amd64 (44.1-1build4) ...
Setting up unity-greeter (24.04.1-0ubuntu1) ...
Setting up gnome-bluetooth-sendto (46.0-1ubuntu1) ...
Setting up gnome-bluetooth (42~3.34.5-13build3) ...
Setting up libgtk-4-media-gstreamer (4.14.5+ds-0ubuntu0.5) ...
Setting up libebbackend-1.2-11t64:amd64 (3.52.3-0ubuntu1) ...
Setting up libecal-2.0-3:amd64 (3.52.3-0ubuntu1) ...
Setting up network-manager-gnome (1.34.0-1ubuntu3) ...
Setting up libedata-cal-2.0-2t64:amd64 (3.52.3-0ubuntu1) ...
Setting up libedata-book-1.2-27t64:amd64 (3.52.3-0ubuntu1) ...
Setting up libedataserverui-1.2-4t64:amd64 (3.52.3-0ubuntu1) ...
Setting up indicator-datetime (15.10+21.04.20210304-0ubuntu6) ...
Setting up unity-control-center (15.04.0+23.04.20230220-0ubuntu8) ...
Setting up libebook-1.2-21t64:amd64 (3.52.3-0ubuntu1) ...
Setting up indicator-bluetooth (0.0.6+17.10.20170605-0ubuntu5) ...
Setting up evolution-data-server (3.52.3-0ubuntu1) ...
Processing triggers for dictionaries-common (1.29.7) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
saira@ubuntu:~$ _

```

4. Install and enable XRDP (Remote Desktop Protocol server):

sudo apt install xrdp -y

```

Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 xrdp amd64 0.9.24-4 [536 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 libpipewire-0.3-modules-xrdp amd64 0.2-2 [20.6 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/universe amd64 pipewire-module-xrdp all 0.2-2 [3,800 B]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 xorgxrdp amd64 1:0.9.19-1 [65.3 kB]
Fetched 715 kB in 6s (118 kB/s)
Selecting previously unselected package libfuse2t64:amd64.
(Reading database ... 176232 files and directories currently installed.)
Preparing to unpack .../libfuse2t64_2.9.9-8.1build1_amd64.deb ...
Unpacking libfuse2t64:amd64 (2.9.9-8.1build1) ...
Selecting previously unselected package xrdp.
Preparing to unpack .../xrdp_0.9.24-4_amd64.deb ...
Unpacking xrdp (0.9.24-4) ...
Selecting previously unselected package libpipewire-0.3-modules-xrdp:amd64.
Preparing to unpack .../libpipewire-0.3-modules-xrdp_0.2-2_amd64.deb ...
Unpacking libpipewire-0.3-modules-xrdp:amd64 (0.2-2) ...
Selecting previously unselected package pipewire-module-xrdp.
Preparing to unpack .../pipewire-module-xrdp_0.2-2_all.deb ...
Unpacking pipewire-module-xrdp (0.2-2) ...
Selecting previously unselected package xorgxrdp.
Preparing to unpack .../xorgxrdp_1%3a0.9.19-1_amd64.deb ...
Unpacking xorgxrdp (1:0.9.19-1) ...
Setting up libpipewire-0.3-modules-xrdp:amd64 (0.2-2) ...
Setting up xorgxrdp (1:0.9.19-1) ...
Setting up libfuse2t64:amd64 (2.9.9-8.1build1) ...
Setting up xrdp (0.9.24-4) ...

Generating 2048 bit rsa key...

ssl_gen_key_xrdp1 ok

saving to /etc/xrdp/rsakeys.ini

Created symlink /etc/systemd/system/multi-user.target.wants/xrdp-sesman.service → /usr/lib/systemd/system/xrdp-sesman.service.
Created symlink /etc/systemd/system/multi-user.target.wants/xrdp.service → /usr/lib/systemd/system/xrdp.service.
Setting up pipewire-module-xrdp (0.2-2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
saira@ubuntu:~$

```

sudo systemctl enable --now xrdp

```

No VM guests are running outdated hypervisor (qemu) binaries on this host.
saira@ubuntu:~$ sudo systemctl enable --now xrdp
Synchronizing state of xrdp.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable xrdp
saira@ubuntu:~$

```

5. Verify XRDP status:

sudo systemctl status xrdp


```
saira@ubuntu:~$ sudo systemctl status xrdp
● xrdp.service - xrdp daemon
   Loaded: loaded (/usr/lib/systemd/system/xrdp.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-11-04 11:08:26 UTC; 1min 21s ago
     Docs: man:xrdp(8)
           man:xrdp.ini(5)
  Main PID: 16061 (xrdp)
    Tasks: 1 (limit: 4545)
   Memory: 872.0K (peak: 1.5M)
      CPU: 50ms
   CGroup: /system.slice/xrdp.service
           └─16061 /usr/sbin/xrdp

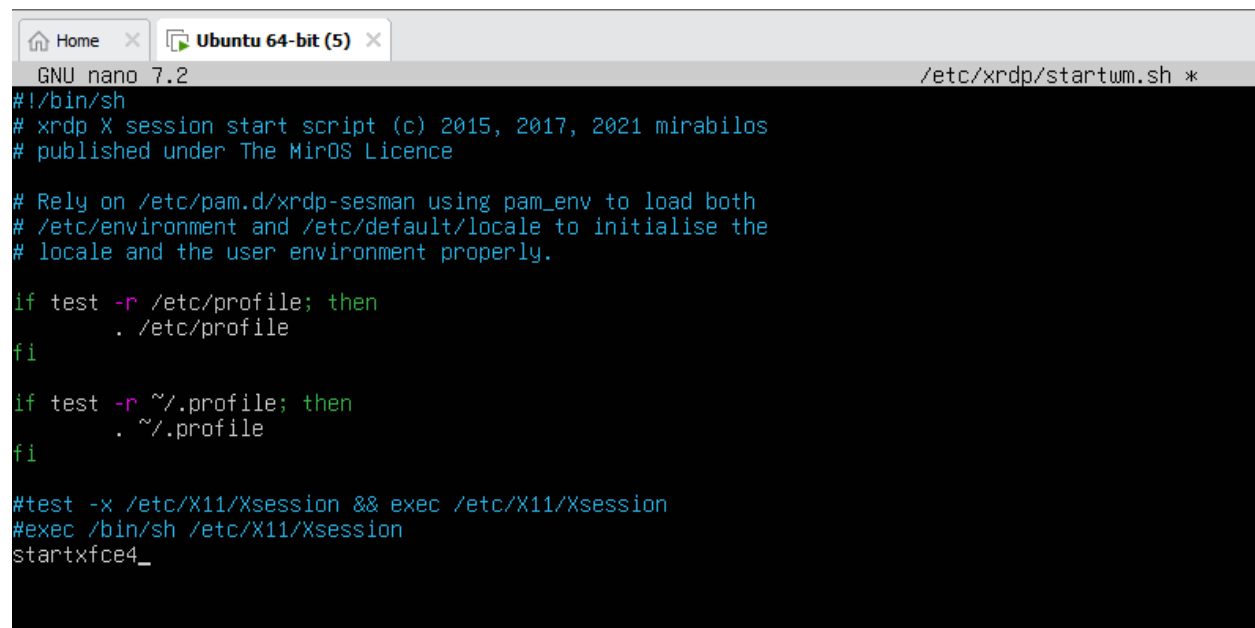
Nov 04 11:08:25 ubuntu systemd[1]: Starting xrdp.service - xrdp daemon...
Nov 04 11:08:25 ubuntu xrdp[16060]: [INFO ] address [0.0.0.0] port [3389] mode 1
Nov 04 11:08:25 ubuntu xrdp[16060]: [INFO ] listening to port 3389 on 0.0.0.0
Nov 04 11:08:25 ubuntu xrdp[16060]: [INFO ] xrdp_listen_pp done
Nov 04 11:08:25 ubuntu systemd[1]: xrdp.service: Can't open PID file /run/xrdp/xrdp.pid (yet?) after start: No such file or directory
Nov 04 11:08:26 ubuntu systemd[1]: Started xrdp.service - xrdp daemon.
Nov 04 11:08:27 ubuntu xrdp[16061]: [INFO ] starting xrdp with pid 16061
Nov 04 11:08:27 ubuntu xrdp[16061]: [INFO ] address [0.0.0.0] port [3389] mode 1
Nov 04 11:08:27 ubuntu xrdp[16061]: [INFO ] listening to port 3389 on 0.0.0.0
Nov 04 11:08:27 ubuntu xrdp[16061]: [INFO ] xrdp_listen_pp done
saira@ubuntu:~$
```

6. Configure XRDp to use XFCE session:

echo xfce4-session > ~/.xsession

7. From a Windows host or RDP client, connect with Remote Desktop (mstsc) to your server IP and login using your Ubuntu username/password. Capture a screenshot of the remote desktop or the RDP session window (if allowed by your environment) and save it as: task5_rdp_connect.png

If you cannot capture a screenshot from the client, show `ss -ltnp` or `ps` evidence that the session was established and save as an alternative.



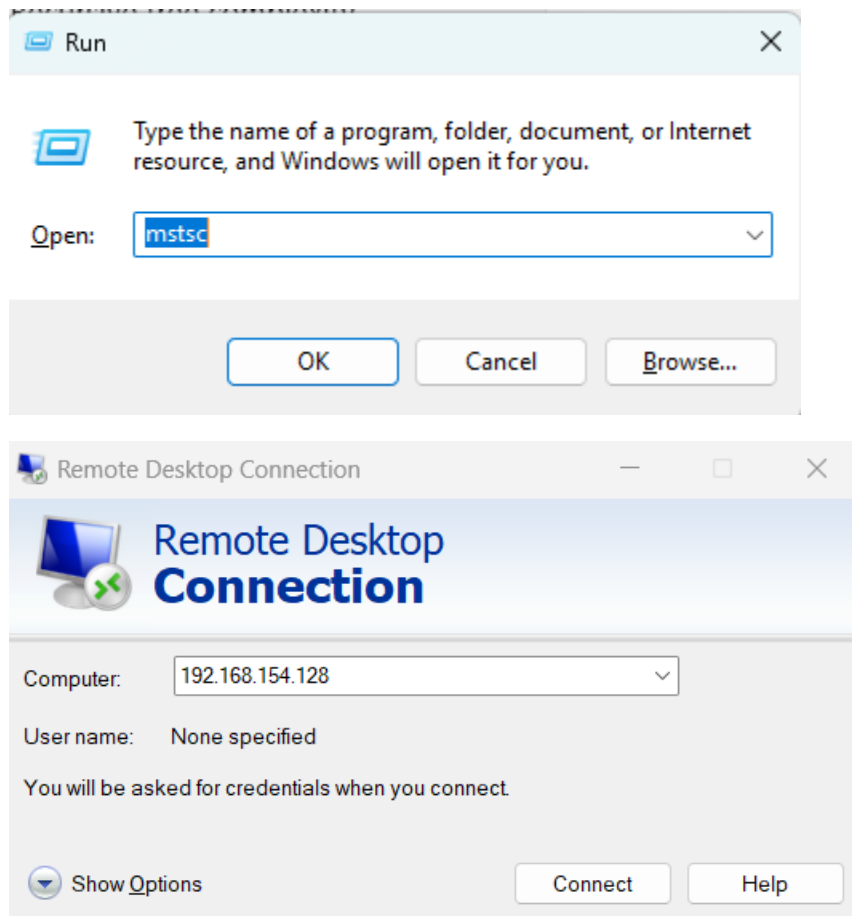
```
GNU nano 7.2 /etc/xrdp/startwm.sh *
#!/bin/sh
# xrdp X session start script (c) 2015, 2017, 2021 mirabilos
# published under The MirOS Licence

# Rely on /etc/pam.d/xrdp-sesman using pam_env to load both
# /etc/environment and /etc/default/locale to initialise the
# locale and the user environment properly.

if test -r /etc/profile; then
    . /etc/profile
fi

if test -r ~/.profile; then
    . ~/.profile
fi

#test -x /etc/X11/Xsession && exec /etc/X11/Xsession
#exec /bin/sh /etc/X11/Xsession
startxfce4_
```



```
saira@ubuntu:~$ ss -ltnp
State      Recv-Q    Send-Q    Local Address:Port    Peer Address:Port    Process
LISTEN     0          4096      127.0.0.53:10:53      0.0.0.0:*
LISTEN     0          4096      127.0.0.1:631        0.0.0.0:*
LISTEN     0          4096      127.0.0.54:53        0.0.0.0:*
LISTEN     0          4096      0.0.0.0:22           0.0.0.0:*
LISTEN     0          2          *:3389               *:3389
LISTEN     0          2          [::1]:3350           [::]:3350
LISTEN     0          4096      [::]:22              [::]:22
LISTEN     0          4096      [::1]:631            [::1]:631
```

After you are in the GUI (local console or RDP session), launch Visual Studio Code (installed in Task 4) from the GUI menu or a terminal inside the GUI. Example command from a GUI terminal

Task 6 - Install lightdm-gtk-greeter and GUI verification - start GUI, open VS Code, take snapshot, then end (GUI)

Steps (inside the host terminal / via SSH)

1. Fix GUI login screen issues (if lightdm / greeter problems appear)

Install LightDM and greeter using Host Terminal:

sudo apt install lightdm lightdm-gtk-greeter -y

```
saira@ubuntu:~$ sudo apt install lightdm lightdm-gtk-greeter -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
lightdm is already the newest version (1.30.0-0ubuntu14).
lightdm set to manually installed.
The following packages were automatically installed and are no longer required:
  ca-certificates-java default-jre-headless fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-jni libgif7 openjdk-21-jre
  openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  gnome-accessibility-themes gnome-themes-extra gnome-themes-extra-data gtk2-engines-pixbuf
The following NEW packages will be installed:
  gnome-accessibility-themes gnome-themes-extra gnome-themes-extra-data gtk2-engines-pixbuf lightdm-gtk-greeter
0 upgraded, 5 newly installed, 0 to remove and 0 not upgraded.
Need to get 2,495 kB of archives.
After this operation, 7,003 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 gnome-accessibility-themes all 3.28-2ubuntu5 [2,296 kB]
1% [1 gnome-accessibility-themes 30.8 kB/2,296 kB 1%]
```

2. Create LightDM config to use XFCE:

```
sudo mkdir -p /etc/lightdm/lightdm.conf.d
```

```
echo -e "[Seat:*]\ngreeter-session=lightdm-gtk-greeter\nuser-session=xfce\nautologin-user-  
timeout=0" | sudo tee /etc/lightdm/lightdm.conf.d/99-xfce.conf
```

```
saira@ubuntu:~$ sudo mkdir -p /etc/lightdm/lightdm.conf.d
saira@ubuntu:~$ echo -e "[Seat:*]\ngreeter-session=lightdm-gtk-greeter\nuser-session=xfce\nautologin-user-  
timeout=0" | sudo tee /etc/lightdm/lightdm.conf.d/99-xfce.conf
[Seat:*]
greeter-session=lightdm-gtk-greeter
user-session=xfce
autologin-user=timeout=0
saira@ubuntu:~$
```

3. Clean up problematic session files and permissions:

```
sudo rm -f /var/lib/lightdm/.Xauthority
```

```
sudo rm -f ~/.Xauthority
```

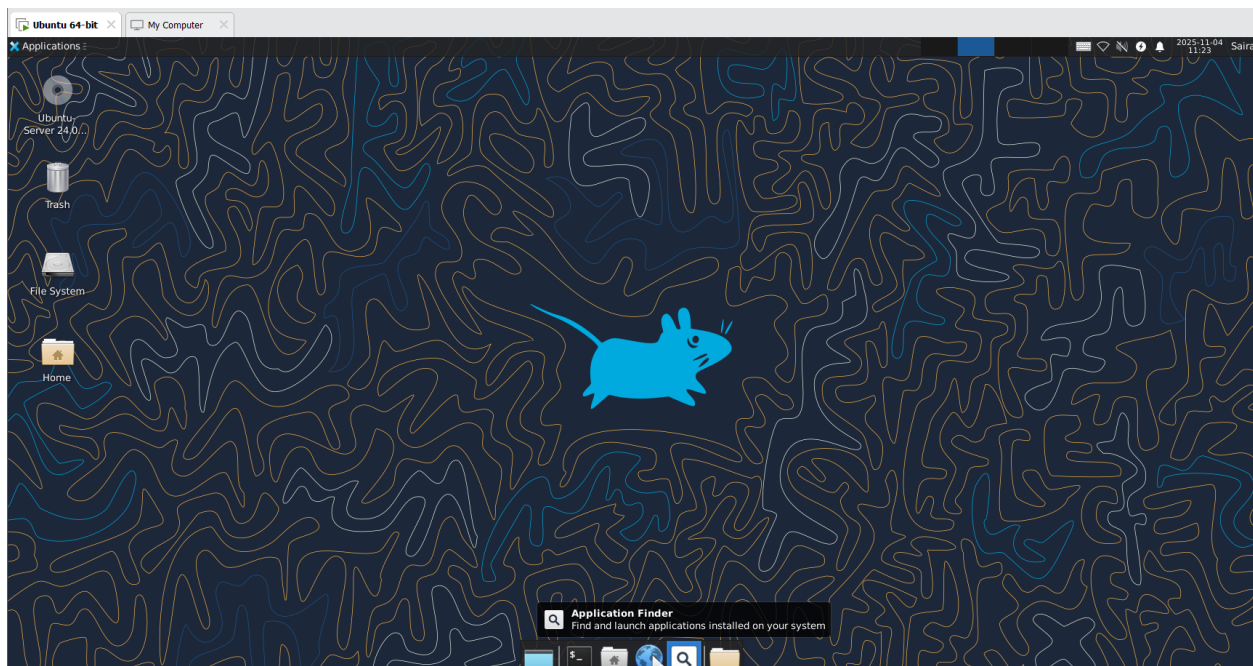
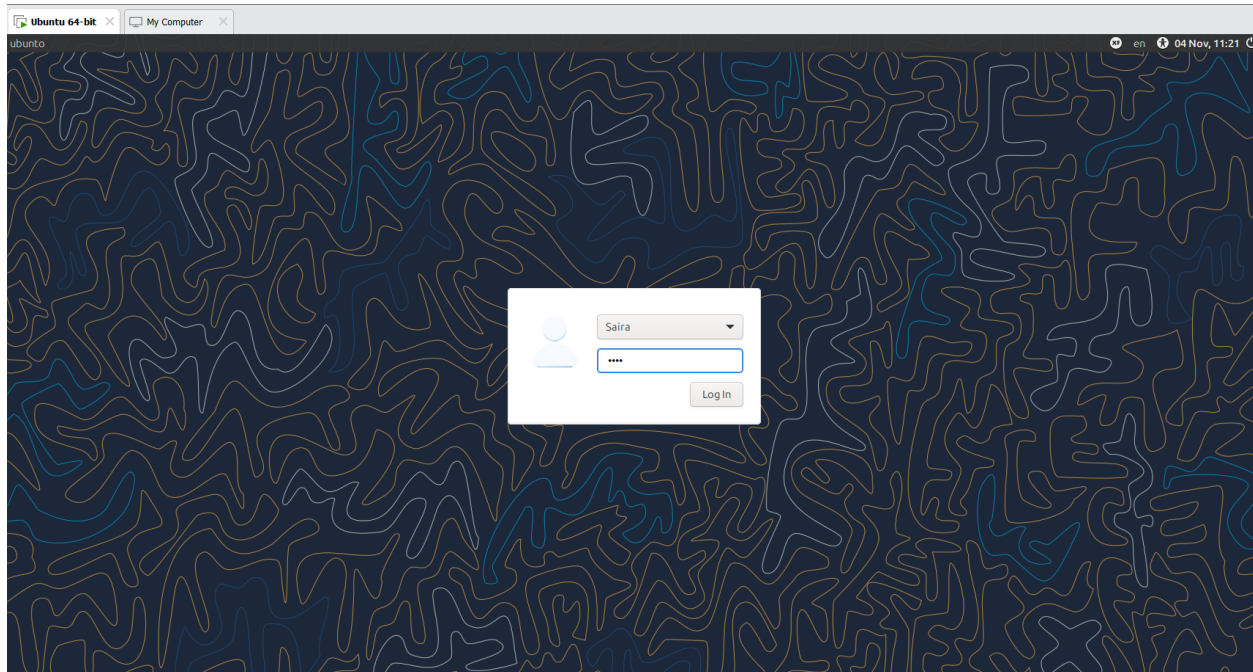
```
sudo rm -rf ~/.cache/sessions
```

```
sudo chown -R $USER:$USER /home/$USER
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
saira@ubuntu:~$ sudo rm -f /var/lib/lightdm/.Xauthority
saira@ubuntu:~$ sudo rm -f ~/.Xauthority
saira@ubuntu:~$ sudo rm -rf ~/.cache/sessions
saira@ubuntu:~$ sudo chown -R saira:saira /home/saira
```

4. Restart LightDM:

```
sudo systemctl restart lightdm
```



5. Control GUI login at boot — ENABLE first, then DISABLE (observe and understand terminal/GUI behavior after each reboot)

Important: students MUST perform the reboot after each target change to observe the boot-time behavior. The sequence below has been adjusted so you ENABLE the GUI boot target first, reboot and observe GUI, then DISABLE the GUI boot target, reboot and observe the CLI.

6. Enable GUI Login Screen (Boot to GUI)
7. Re-enable LightDM and set the graphical target as default:

sudo systemctl enable lightdm

sudo systemctl set-default graphical.target

```
saira@ubuntu:~$ sudo systemctl enable lightdm
Synchronizing state of lightdm.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable lightdm
The unit files have no installation config (WantedBy=, RequiredBy=, UpheldBy=,
Also=, or Alias= settings in the [Install] section, and DefaultInstance= for
template units). This means they are not meant to be enabled or disabled using systemctl.
```

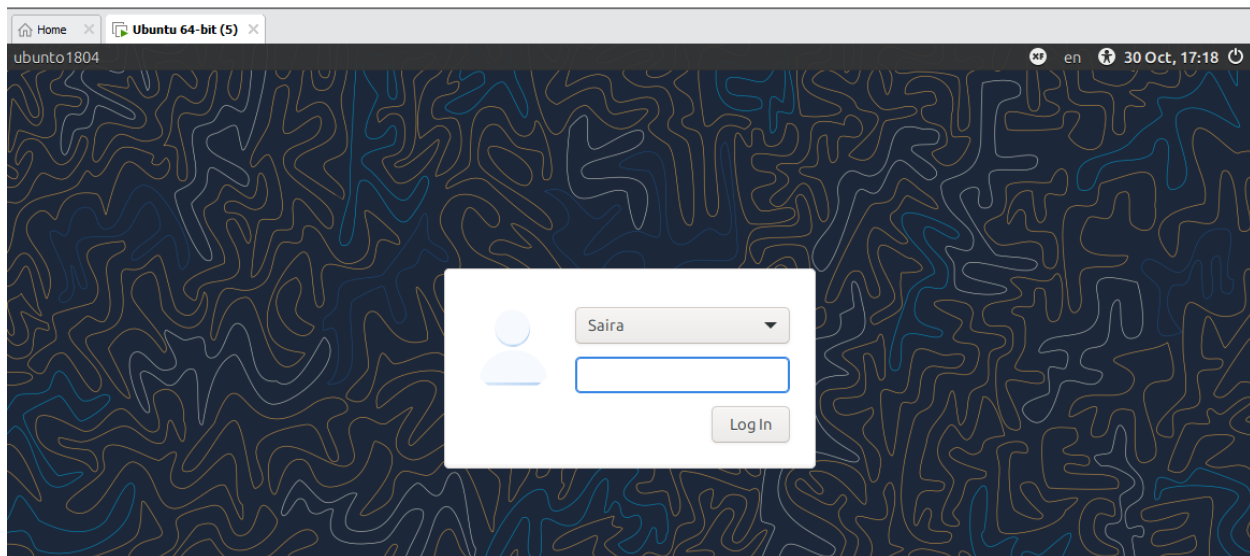
Possible reasons for having these kinds of units are:

- A unit may be statically enabled by being symlinked from another unit's .wants/, .requires/, or .upholds/ directory.
- A unit's purpose may be to act as a helper for some other unit which has a requirement dependency on it.
- A unit may be started when needed via activation (socket, path, timer, D-Bus, udev, scripted systemctl call, ...).
- In case of template units, the unit is meant to be enabled with some instance name specified.

```
saira@ubuntu:~$ sudo systemctl set-default graphical.target
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/graphical.target.
saira@ubuntu:~$
```

8. Reboot the VM to observe that it boots to the GUI login screen:

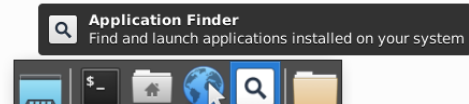
sudo reboot



9. Disable GUI Login Screen (Boot to CLI)

```
saira@ubuntu:~$ sudo systemctl set-default graphical.target
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/graphical.target.
saira@ubuntu:~$ sudo systemctl set-default multi-user.target
Removed "/etc/systemd/system/default.target".
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/multi-user.target.
saira@ubuntu:~$ sudo systemctl disable lightdm
Synchronizing state of lightdm.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable lightdm
The unit files have no installation config (WantedBy=, RequiredBy=, UpheldBy=,
Also=, or Alias= settings in the [Install] section, and DefaultInstance= for
template units). This means they are not meant to be enabled or disabled using systemctl.

Possible reasons for having these kinds of units are:
• A unit may be statically enabled by being symlinked from another unit's
.wants/, .requires/, or .upholds/ directory.
• A unit's purpose may be to act as a helper for some other unit which has
a requirement dependency on it.
• A unit may be started when needed via activation (socket, path, timer,
D-Bus, udev, scripted systemctl call, ...).
• In case of template units, the unit is meant to be enabled with some
instance name specified.
saira@ubuntu:~$ sudo systemctl stop lightdm
```



10. Set the default boot target to multi-user (text mode) and disable LightDM so the system boots to the terminal:

sudo systemctl set-default multi-user.target

```
saira@ubuntu1804:~$ sudo systemctl set-default multi-user.target
[sudo] password for saira:
Removed "/etc/systemd/system/default.target".
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/multi-user.target
```

sudo systemctl disable lightdm

```
saira@ubuntu1804:~$ sudo systemctl disable lightdm
Synchronizing state of lightdm.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable lightdm
Removed "/etc/systemd/system/display-manager.service".
saira@ubuntu1804:~$
```

```
saira@ubuntu:~$ sudo systemctl set-default multi-user.target
[sudo] password for saira:
saira@ubuntu:~$ sudo systemctl disable lightdm
Synchronizing state of lightdm.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable lightdm
The unit files have no installation config (WantedBy=, RequiredBy=, UpheldBy=,
Also=, or Alias= settings in the [Install] section, and DefaultInstance= for
template units). This means they are not meant to be enabled or disabled using systemctl.

Possible reasons for having these kinds of units are:
• A unit may be statically enabled by being symlinked from another unit's
.wants/, .requires/, or .upholds/ directory.
• A unit's purpose may be to act as a helper for some other unit which has
a requirement dependency on it.
• A unit may be started when needed via activation (socket, path, timer,
D-Bus, udev, scripted systemctl call, ...).
• In case of template units, the unit is meant to be enabled with some
instance name specified.
saira@ubuntu:~$
```

11. Reboot the VM to observe that it boots to the terminal (CLI):

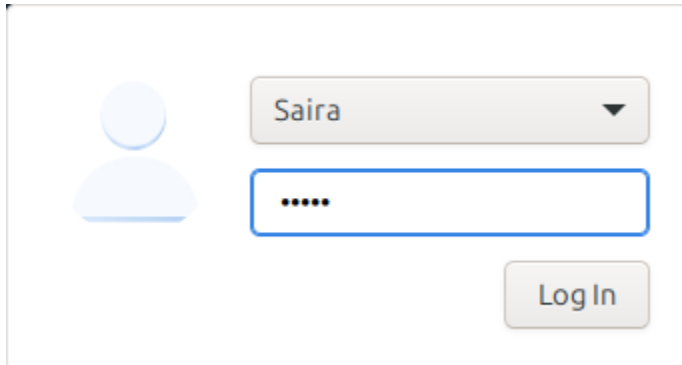
sudo reboot

```
saira@ubuntu1804:~$ sudo reboot
```

12. Start/Stop GUI manually (no reboot)

You can start the GUI session without changing the boot target. This is useful if you want to keep the boot target as CLI but run GUI temporarily:

sudo systemctl start lightdm # start GUI now



13. Press Ctrl + Alt + F3 to switch back to TTY. You can stop the GUI session without changing the boot target.

sudo systemctl stop lightdm # stop GUI now



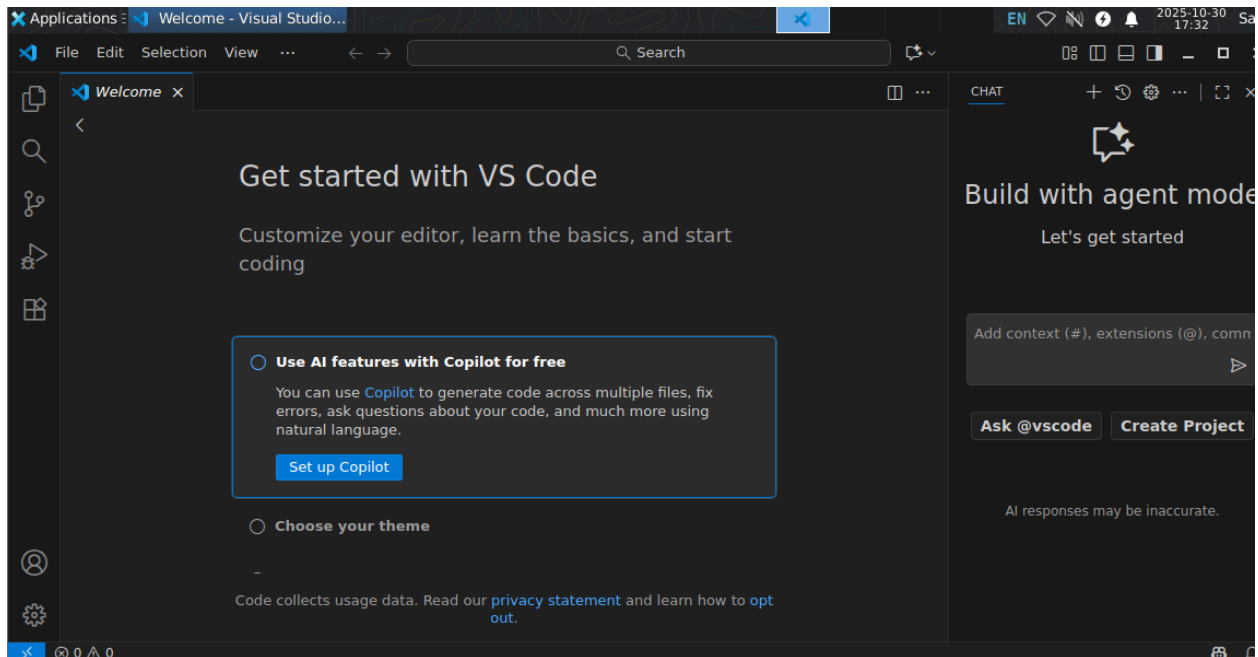
14. Start the GUI (if the system is currently set to CLI or the GUI is not running):

sudo systemctl start lightdm

```
saira@ubuntu1804:~$ sudo systemctl start lightdm
[sudo] password for saira: _
```


15. In the GUI session launch Visual Studio Code (installed earlier in Task 4). From a GUI terminal inside the desktop, run:

Code



Task 7 - Install Google Chrome by adding its apt source & key (Chrome)

Steps (inside the VM terminal or GUI terminal or host terminal / via SSH)

1. (Learning step — first command must be the install attempt) Attempt to install Google Chrome directly to see the failure when the repo/key are missing:

sudo apt install google-chrome-stable -y

```
saira@ubuntu:~$ sudo apt install google-chrome-stable -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  ca-certificates-java default-jre-headless fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-jni libglib7 openjdk-21-jre
  openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  google-chrome-stable
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 118 MB of archives.
After this operation, 390 MB of additional disk space will be used.
Get:1 http://dl.google.com/linux/chrome/deb stable/main amd64 google-chrome-stable amd64 142.0.7444.59-1 [118 MB]
1% [1 google-chrome-stable 1,589 kB/118 MB 1%]
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
saira@ubuntu:~$ google-chrome --version
Google Chrome 142.0.7444.59
saira@ubuntu:~$
```

Expected: this will typically fail with "Unable to locate package google-chrome-stable" or similar because the Google repo/key are not yet configured.

2. Inspect apt configuration so you understand why install failed. List the /etc/apt directory:

ls -la /etc/apt

```
saira@ubuntu:~$ ls -la /etc/apt
total 52
drwxr-xr-x  9 root root  4096 Nov  4 11:52 .
drwxr-xr-x 142 root root 12288 Nov  4 11:15 ..
drwxr-xr-x  2 root root  4096 Nov  4 09:47 apt.conf.d
drwxr-xr-x  2 root root  4096 Mar 31 2024 auth.conf.d
drwxr-xr-x  2 root root  4096 Nov  4 11:42 keyrings
drwxr-xr-x  2 root root  4096 Aug  5 17:14 preferences.d
drwxr-xr-x  2 root root  4096 Aug  5 17:14 preferences.d.save
-rw-r--r--  1 root root   347 Nov  4 11:52 sources.list
-rw-r--r--  1 root root    70 Nov  4 11:52 sources.list.backup
drwxr-xr-x  2 root root  4096 Nov  4 11:57 sources.list.d
drwxr-xr-x  2 root root  4096 Nov  4 11:57 trusted.gpg.d
saira@ubuntu:~$ _
```

3. View the main /etc/apt/sources.list:

cat /etc/apt/sources.list

```
saira@ubuntu:~$ cat /etc/apt/sources.list
deb http://archive.ubuntu.com/ubuntu noble main restricted universe multiverse

deb http://archive.ubuntu.com/ubuntu noble-updates main restricted universe multiverse

deb http://archive.ubuntu.com/ubuntu noble-backports main restricted universe multiverse

deb http://security.ubuntu.com/ubuntu noble-security main restricted universe multiverse
saira@ubuntu:~$ _
```

4. List files under /etc/apt/sources.list.d:

ls -la /etc/apt/sources.list.d/

```
saira@ubuntu:~$ ls -la /etc/apt/sources.list.d/
total 20
drwxr-xr-x 2 root root 4096 Nov  4 11:57 .
drwxr-xr-x 9 root root 4096 Nov  4 11:52 ..
-rw-r--r-- 1 root root  190 Nov  4 11:57 google-chrome.list
-rw-r--r-- 1 root root  152 Nov  4 11:41 ubuntu.sources
-rw-r--r-- 1 root root 2552 Aug  5 17:02 ubuntu.sources.curtin.orig
saira@ubuntu:~$
```

5. If there is a file named ubuntu.sources (or similarly named source file), display it to see whether Chrome's repo is present:

cat /etc/apt/sources.list.d/ubuntu.sources

```
saira@ubuntu:~$ cat /etc/apt/sources.list.d/ubuntu.sources
Types: deb
URIs: http://dl.google.com/linux/chrome/deb/
Suites: stable
Components: main
Architectures: amd64
Signed-By: /etc/apt/keyrings/google.gpg

saira@ubuntu:~$ _
```

6. Add Chrome repository metadata to a sources file (method A — using ubuntu.sources). Open or create the file and append the stanza (you can alternatively use the preferred one-line method in step 11):

sudo nano /etc/apt/sources.list.d/ubuntu.sources

Append these exact lines at the end of the file:

Types: deb

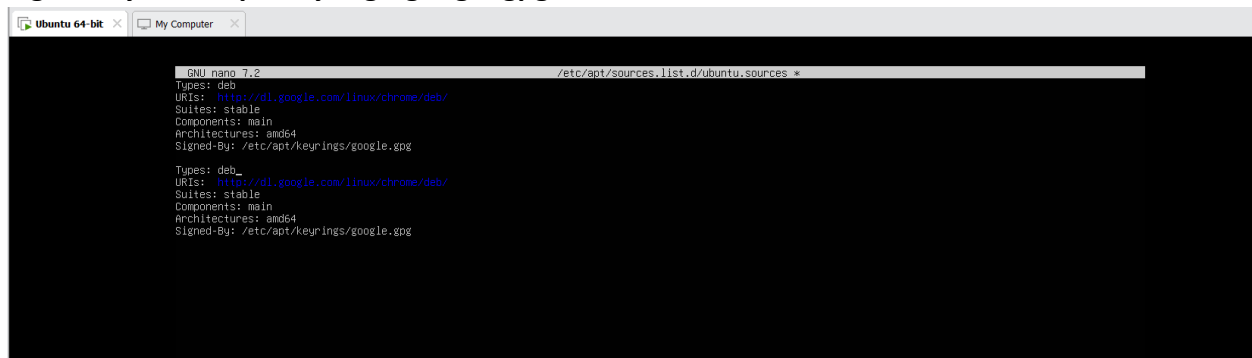
URIs: http://dl.google.com/linux/chrome/deb/

Suites: stable

Components: main

Architectures: amd64

Signed-By: /etc/apt/keyrings/google.gpg



```
GNU nano 7.2 /etc/apt/sources.list.d/ubuntu.sources *
Types: deb
URIs: http://dl.google.com/linux/chrome/deb/
Suites: stable
Components: main
Architectures: amd64
Signed-By: /etc/apt/keyrings/google.gpg

Types: deb
URIs: http://dl.google.com/linux/chrome/deb/
Suites: stable
Components: main
Architectures: amd64
Signed-By: /etc/apt/keyrings/google.gpg
```

7. Ensure the keyrings directory exists and import Google's signing key:

curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /etc/apt/keyrings/google.gpg

```
saira@ubuntu:~$ curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /etc/apt/keyrings/google.gpg
File '/etc/apt/keyrings/google.gpg' exists. Overwrite? (y/N) y
saira@ubuntu:~$
```

8. Update apt and attempt to install Chrome again (now that repo + key are added):

sudo apt update

```
.sources:1
M: Target CNF (main/cnf/Commands-amd64) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target CNF (main/cnf/Commands-all) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target Packages (main/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
M: Target Packages (main/binary-all/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
M: Target Translations (main/i18n/Translation-en_GB) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
M: Target Translations (main/i18n/Translation-en) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
M: Target DEP-11 (main/dep11/Components-amd64.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
M: Target DEP-11 (main/dep11/Components-all.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
M: Target CNF (main/cnf/Commands-amd64) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
M: Target CNF (main/cnf/Commands-all) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
M: Target Packages (main/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target Packages (main/binary-all/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target Translations (main/i18n/Translation-en_GB) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target Translations (main/i18n/Translation-en) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target DEP-11 (main/dep11/Components-amd64.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target DEP-11 (main/dep11/Components-all.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target CNF (main/cnf/Commands-amd64) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target CNF (main/cnf/Commands-all) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target Packages (main/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
M: Target Packages (main/binary-all/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
M: Target Translations (main/i18n/Translation-en_GB) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
M: Target Translations (main/i18n/Translation-en) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
M: Target DEP-11 (main/dep11/Components-amd64.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
M: Target DEP-11 (main/dep11/Components-all.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
M: Target CNF (main/cnf/Commands-amd64) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
M: Target CNF (main/cnf/Commands-all) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:3 and /etc/apt/sources.list.d/ubuntu.sources:2
saina@ubuntu:~$ _
```

sudo apt install google-chrome-stable -y

```
saina@ubuntu:~$ sudo apt install google-chrome-stable -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
google-chrome-stable is already the newest version (142.0.7444.59-1).
The following packages were automatically installed and are no longer required:
  ca-certificates-java default-jre-headless fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-jni libgif7 openjdk-21-jre
  openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
saina@ubuntu:~$
```

9. Alternate (preferred, cleaner) method — create a single google-chrome.list entry

Cleanup before alternate method you added the chrome earlier and want to switch to the preferred method:

sudo apt remove google-chrome-stable -y

```
saira@ubuntu:~$ sudo apt install google-chrome-stable -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
google-chrome-stable is already the newest version (142.0.7444.59-1).
The following packages were automatically installed and are no longer required:
  ca-certificates-java default-jre-headless fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-jni libgif7 openjdk-21-jre
  openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
saira@ubuntu:~$ sudo apt remove google-chrome-stable -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  ca-certificates-java default-jre-headless fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-jni libgif7 openjdk-21-jre
  openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  google-chrome-stable
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
After this operation, 390 MB disk space will be freed.
(Reading database ... 180722 files and directories currently installed.)
Removing google-chrome-stable (142.0.7444.59-1) ...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu5) ...
Rebuilding /usr/share/applications/bamf-2.index...
saira@ubuntu:~$
```

sudo nano /etc/apt/sources.list.d/ubuntu.sources # remove the chrome stanza you added earlier, save and exit

```
GNU nano 7.2 /etc/apt/sources.list.d/ubuntu.sources *
Types: deb
URIs: http://dl.google.com/linux/chrome/deb/
Suites: stable
Components: main
Architectures: amd64
Signed-By: /etc/apt/keyrings/google.gpg
```

sudo rm -f /etc/apt/keyrings/google.gpg

10. Create a dedicated one-line list file for Google Chrome (preferred):

**echo "deb [arch=amd64 signed-by=/etc/apt/keyrings/google.gpg]
http://dl.google.com/linux/chrome/deb/ stable main" | sudo tee
/etc/apt/sources.list.d/google-chrome.list > /dev/null**

```
saira@ubuntu:~$ echo "deb [arch=amd64 signed-by=/etc/apt/keyrings/google.gpg] http://dl.google.com/linux/chrome/deb/ stable main" | sudo tee /etc/apt/sources.li
st.d/google-chrome.list > /dev/null
saira@ubuntu:~$
saira@ubuntu:~$ _
```

11. Verify the new file exists:

ls -la /etc/apt/sources.list.d/

```
saira@ubuntu:~$ ls -la /etc/apt/sources.list.d/
total 20
drwxr-xr-x 2 root root 4096 Nov  4 12:06 .
drwxr-xr-x 9 root root 4096 Nov  4 11:52 ..
-rw-r--r-- 1 root root 107 Nov  4 12:07 google-chrome.list
-rw-r--r-- 1 root root 150 Nov  4 12:06 ubuntu.sources
-rw-r--r-- 1 root root 2552 Aug  5 17:02 ubuntu.sources.curtin.orig
saira@ubuntu:~$
```

12. Re-add the Google signing key (if removed previously or not present):

sudo mkdir -p /etc/apt/keyrings

```
saira@ubuntu:~$ sudo mkdir -p /etc/apt/keyrings
saira@ubuntu:~$ curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /etc/apt/keyrings/google.gpg
```

curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /etc/apt/keyrings/google.gpg

13. Update apt and install Chrome (preferred flow):

sudo apt update

```
saira@ubuntu:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
N: Target Packages (main/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
N: Target Packages (main/binary-all/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
N: Target Translations (main/i18n/Translation-en_GB) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
N: Target Translations (main/i18n/Translation-en) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
N: Target DEP-11 (main/dep11/Components-amd64.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
N: Target DEP-11 (main/dep11/Components-all.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
N: Target CNF (main/cnf/Commands-amd64) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
N: Target CNF (main/cnf/Commands-all) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
N: Target Packages (main/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
N: Target Packages (main/binary-all/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
N: Target Translations (main/i18n/Translation-en_GB) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
N: Target Translations (main/i18n/Translation-en) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
N: Target DEP-11 (main/dep11/Components-amd64.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
N: Target DEP-11 (main/dep11/Components-all.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
N: Target CNF (main/cnf/Commands-amd64) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
N: Target CNF (main/cnf/Commands-all) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
saira@ubuntu:~$
```

sudo apt install google-chrome-stable -y

```
saira@ubuntu:~$ sudo apt install google-chrome-stable -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  ca-certificates-java default-jre-headless fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-jni libgif7 openjdk-21-jre
  openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  google-chrome-stable
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 118 MB of archives.
After this operation, 390 MB of additional disk space will be used.
Get:1 https://dl.google.com/linux/chrome/deb stable/main amd64 google-chrome-stable amd64 142.0.7444.59-1 [118 MB]
2% [1 google-chrome-stable 2,528 kB/118 MB 2%]
```

Task 8 - Install applications via PPA (Audacity & OBS) and launch

Goal: Add PPAs for Audacity and OBS Studio, update apt, install the packages, and launch them (or verify installation) from the CLI/GUI. Capture screenshots for each step.

Steps (inside the VM terminal or GUI terminal)

1. Add the Audacity PPA, update apt and install audacity:

sudo add-apt-repository ppa:ubuntuhandbook1/audacity -y

```
If the packages here are helpful, you may buy me a coffee:
    https://ko-fi.com/ubuntuhandbook1
More info: https://launchpad.net/~Ubuntuhandbook1/+archive/ubuntu/audacity
Adding repository.
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:6 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease [18.1 kB]
Get:7 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble/main amd64 Packages [1,064 B]
Get:8 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble/main Translation-en [492 B]
Fetched 19.6 kB in 3s (7,253 B/s)
Reading package lists... Done
W: Target Packages (main/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target Packages (main/binary-all/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target Translations (main/i18n/Translation-en_GB) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target Translations (main/i18n/Translation-en) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target DEP-11 (main/dep11/Components-amd64.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target DEP-11 (main/dep11/Components-all.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target CNF (main/cnf/Commands-amd64) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target CNF (main/cnf/Commands-all) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target Packages (main/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target Packages (main/binary-all/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target Translations (main/i18n/Translation-en_GB) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target Translations (main/i18n/Translation-en) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target DEP-11 (main/dep11/Components-amd64.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target DEP-11 (main/dep11/Components-all.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target CNF (main/cnf/Commands-amd64) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target CNF (main/cnf/Commands-all) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
sair@ubuntu:~$ _
```

sudo apt update

```
W: Target Packages (main/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
sair@ubuntu:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:5 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:6 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
W: Target Packages (main/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target Packages (main/binary-all/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target Translations (main/i18n/Translation-en_GB) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target Translations (main/i18n/Translation-en) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target DEP-11 (main/dep11/Components-amd64.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target DEP-11 (main/dep11/Components-all.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target CNF (main/cnf/Commands-amd64) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target CNF (main/cnf/Commands-all) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target Packages (main/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target Packages (main/binary-all/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target Translations (main/i18n/Translation-en_GB) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target Translations (main/i18n/Translation-en) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target DEP-11 (main/dep11/Components-amd64.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target DEP-11 (main/dep11/Components-all.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target CNF (main/cnf/Commands-amd64) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
W: Target CNF (main/cnf/Commands-all) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
sair@ubuntu:~$
```

sudo apt install audacity -y

```

Setting up libqt5qmlmodels5:amd64 (5.15.13+dfsg-1ubuntu0.1) ...
Setting up libqt5widgets5t64:amd64 (5.15.13+dfsg-1ubuntu1) ...
Setting up libqt5svg5:amd64 (5.15.13-1) ...
Setting up audacity (3.7.5-0build1~ubuntu24.04) ...
Setting up qt5-gtk-platformtheme:amd64 (5.15.13+dfsg-1ubuntu1) ...
Setting up libqt5waylandclient5:amd64 (5.15.13-1) ...
Setting up libqt5quick5:amd64 (5.15.13+dfsg-1ubuntu0.1) ...
Setting up libqt5waylandcompositor5:amd64 (5.15.13-1) ...
Setting up qtwayland5:amd64 (5.15.13-1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for shared-mime-info (2.4-4) ...
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu5) ...
Rebuilding /usr/share/applications/bamf-2.index...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for gnome-menus (3.36.0-1.1ubuntu3) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
saira@ubuntu:~$ _

```

2. Launch Audacity (from GUI or CLI). On a headless server you may not get a GUI window — if you are using the XFCE GUI session, launch from a GUI terminal or run check for binary:

audacity --version

audacity

```

saira@ubuntu:~$ audacity --version
saira@ubuntu:~$ audacity
saira@ubuntu:~$

```

3. Add the OBS Studio PPA, update apt and install obs-studio:

sudo add-apt-repository ppa:obsproject/obs-studio -y

```
saira@ubuntu:~$ sudo add-apt-repository ppa:obsproject/obs-studio -y
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu/
Suites: noble
Components: main
'

Description:
Latest stable release of OBS Studio
More info: https://launchpad.net/~obsproject/+archive/ubuntu/obs-studio
Adding repository.
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:5 http://dl.google.com/linux/chrome/deb stable InRelease
Get:6 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease [17.8 kB]
Hit:7 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Get:8 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble/main amd64 Packages [1,172 B]
Get:9 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble/main Translation-en [160 B]
Fetched 19.1 kB in 2s (8,921 B/s)
Reading package lists... 6%
```

sudo apt update

```
M: Target CNF (main/cnf/Commands-all) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target Packages (main/binary-amd64/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target Packages (main/binary-all/Packages) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target Translations (main/i18n/Translation-en_GB) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target Translations (main/i18n/Translation-en) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target DEP-11 (main/dep11/Components-amd64.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target DEP-11 (main/dep11/Components-all.yml) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target CNF (main/cnf/Commands-amd64) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
M: Target CNF (main/cnf/Commands-all) is configured multiple times in /etc/apt/sources.list.d/google-chrome.list:1 and /etc/apt/sources.list.d/ubuntu.sources:1
saira@ubuntu:~$
```

sudo apt install obs-studio -y

```
Get:73 http://archive.ubuntu.com/ubuntu noble/universe amd64 qt6-gtk-platformtheme amd64 6.4.2+dfsg-21.1build5 [55.1 kB]
Get:74 http://archive.ubuntu.com/ubuntu noble/universe amd64 qt6-translations-l10n all 6.4.2-1 [1,761 kB]
Get:75 http://archive.ubuntu.com/ubuntu noble/universe amd64 pocketsphinx-en-us all 0.8.0+real5prealpha+1-15ubuntu5 [27.4 MB]
62% [2 obs-studio 19.8 MB/105 MB 19%]
```

4. Launch OBS Studio (from GUI or verify binary presence):

obs --version

obs

```
saira@ubuntu:~$ obs --version
OBS Studio - 32.0.2
saira@ubuntu:~$ obs
qt.qpa.xcb: could not connect to display
qt.qpa.plugin: Could not load the Qt platform plugin "xcb" in "" even though it was found.
This application failed to start because no Qt platform plugin could be initialized. Reinstalling the application may fix this problem.

Available platform plugins are: vnc, minimal, offscreen, vkhrdisplay, wayland, linuxfb, xcb, eglfs, wayland-egl, minimalegl.

Aborted (core dumped)
saira@ubuntu:~$
```

Task 9 - Create a Kubernetes sample YAML using vim

Goal: Confirm vim availability, create a working directory for Lab5, and use vim to create and save a Kubernetes Pod manifest named k8s-sample.yaml.

Steps (inside the VM terminal or host terminal / via SSH)

1. Check whether vim is installed by running:

vim

- If vim opens, you will see the vim interface; if not installed you will see a command-not-found error.

```
VIM - Vi IMproved

        version 9.1.697
        by Bram Moolenaar et al.
        Modified by team+vim@tracker.debian.org
        Vim is open source and freely distributable


        Become a registered Vim user!
type  :help register<Enter>    for information

type  :q<Enter>                to exit
type  :help<Enter> or <F1>    for on-line help
type  :help version9<Enter>   for version info
```

- If vim is not installed, install it:

sudo apt update

sudo apt install vim -y

2. Create the Lab5 working directory in your home and change into it:

mkdir -p ~/Lab5

cd ~/Lab5

```
saira@ubuntu:~$ mkdir -p ~/Lab5
saira@ubuntu:~$ cd ~/Lab5
saira@ubuntu:~/Lab5$
```

3. Create the Kubernetes sample file using vim:

vim k8s-sample.yaml

- Once vim opens, enable insert mode by pressing i, then paste the following YAML exactly:

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
      restartPolicy: Always
```

4. Exit insert mode by pressing Esc, then save and quit vim with:

`:wq`

```
~
:wq_
```

Task 10 - Edit the Kubernetes YAML - add annotation, verify, then discard temporary change

Steps (inside the VM terminal or host terminal / via SSH)

1. Open the manifest with vim:

`cd ~/Lab5`

`vim k8s-sample.yaml`

```
apiVersion: v1  
kind: Pod  
metadata:  
  name: nginx-pod  
spec:  
  containers:  
    - name: nginx  
      image: nginx:1.19  
      ports:  
        - containerPort: 80  
      restartPolicy: Always
```

"k8s-sample.yaml" 21L, 294B

2. Add the annotation under the metadata section (indentation must match YAML).
Example (insert these lines under metadata:):

annotations:

lab: lesson11

- After adding the lines, press Esc to return to command mode and save & quit:

:wq

```
~
~
:wq_
```

Example verify command:

```
cat k8s-sample.yaml
```

```

k8s-sample.yaml: File, 805B, written
saira@ubuntu:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1

kind: Pod

metadata:
  lab: lesson11
  name: nginx-pod

spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
      restartPolicy: Always
saira@ubuntu:~/Lab5$ _

```

3. Discard changes (practice: make a temporary edit and exit without saving)

- Open the file again:

vim k8s-sample.yaml

- Enter insert mode (i) and add a temporary comment line anywhere, for example:

temp: do-not-keep

```

apiVersion: v1

kind: Pod

metadata:
  lab: lesson11
  name: nginx-pod

spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
      restartPolicy: Always
      # temp: do-not-keep_
~
~
~

```

- Do NOT save. Press Esc to go back to command mode, then force quit without saving:

:q!

- Verify the file does NOT contain the temporary comment:

cat k8s-sample.yaml

```
saira@ubuntu:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1

kind: Pod

metadata:
  lab: lesson11
  name: nginx-pod

spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
      restartPolicy: Always
saira@ubuntu:~/Lab5$
```

Task 11 - Vim editing practice - delete, undo, numeric deletes, navigation

Steps (inside the VM terminal or host terminal / via SSH)

1. Open the file with vim:

cd ~/Lab5

vim k8s-sample.yaml

```

apiVersion: v1
kind: Pod
metadata:
  lab: lesson11
  name: nginx-pod

  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always
~
~
~

```

2. Delete the line containing image: nginx:1.19 with a single command:

- Ensure you are in command mode (press Esc), move the cursor to the line with image: nginx:1.19, then delete that line with:

dd

```

apiVersion: v1
kind: Pod
metadata:
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always
~
~

```

- Immediately undo the deletion with:

u

```

apiVersion: v1
kind: Pod
metadata:
  lab: lesson11
  name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always
~
~
~
~

```

3. Delete 3 lines at once using the numeric delete command:

- In command mode, position the cursor on the first line of the three you want to delete (for example start at the image: line again), then run:

d3d

(or 3dd or d3<enter> — either numeric prefix form is acceptable; document which you used)

```

apiVersion: v1
kind: Pod
metadata:
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always
~

```

- Immediately undo the deletion with:

u

```

apiVersion: v1
kind: Pod
metadata:
  lab: lesson11
  name: nginx-pod

  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always
~

```

4. Navigation practice (from command mode)

- Jump to the first line:

1G

```
apiVersion: v1
```

- Jump to the last line:

G

```
restartPolicy: Always
```

5. Exit vim (no changes should remain if you undid properly):

:q

```
~
:q!
```

Task 12 - Vim search, add matches, substitute, undo.

Steps (inside the VM terminal / in the ~/Lab5 directory)

1. Open the file with vim:

cd ~/Lab5

vim k8s-sample.yaml


```

apiVersion: v1
kind: Pod
metadata:
  lab: lesson11
  name: nginx-pod

  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always
~
~
~

```

2. Search for the string nginx using the forward search command:

- From command mode type:

/nginx

```

apiVersion: v1
kind: Pod
metadata:
  lab: lesson11
  name: nginx-pod

```

3. Move to the next match and previous match:

- Press n to jump to the next match (capture screenshot if desired) and press N to jump back to the previous match.

```
image: nginx:1.19
```

```
- name: nginx
```

4. Add two more occurrences of the word nginx in the file:

- Enter insert mode (i) at an appropriate place (for example add two new comment lines or add them under metadata as comments), type two new occurrences of nginx, then press Esc to return to command mode and save the file with :w.

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    lab: lesson11
    name: nginx-pod
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always
```

5. Demonstrate that n cycles forward through all matches:

- In vim command mode press `/nginx` Enter, then repeatedly press `n` to cycle forward through each match.


```

apiVersion: v1
kind: Pod
metadata:
  lab: lesson11
  name: webapp-pod

  spec:
    containers:
      - name: webapp
        webapp
        webapp
        image: webapp:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always

```

7. Immediately undo the substitution using u:

- Press u in command mode to undo the last change.

```

apiVersion: v1
kind: Pod
metadata:
  lab: lesson11
  name: nginx-pod

  spec:
    containers:
      - name: nginx
        nginx
        nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
        restartPolicy: Always

```

8. Quit vim without saving any accidental changes (if you didn't already save after substitution and want to discard):

- If you want to ensure no changes were written, exit with:

:q!

- If you intentionally saved the substitution and then undid it and wish to quit normally, use :q